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Display device, display method, and computer-readable storage medium

Abstract

A display device includes a display unit; a face angle detector configured to detect a face angle that is an angle between the display unit and a face of a user; and a display controller configured to adjust an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of International Application No. PCT/JP2023/006109, filed on Feb. 21, 2023, which designates the United States, incorporated herein by reference, and which claims the benefit of priority from Japanese Patent Application Nos. 2022-031816 and 2022-031817 each filed on Mar. 2, 2022, the entire contents of both of which are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to a display device, a display method, and a computer-readable storage medium.

(2) A display device that rotates a subject of display according to an angle between a display unit and the face of a user has been known (for example, refer to Japanese Laid-open Patent Publication No. 2015-167010).

(3) In the display device described in Patent Literature 1, when a subject of display is rotated, a part that a user is looking at right before the rotation sometimes is not displayed in an appropriate position. Furthermore, the subject of display sometimes rotates at the time when the user is not looking at the display unit, for example, in the case where the user has his/her eyes closed, the case where the user is looking away from the display unit, or the like. In this case, there is a possibility that the user may be given with uncomfortableness. For this reason, a configuration for performing an operation of rotating the subject of display appropriately is required.

SUMMARY

(4) A display device according to the present disclosure includes: a display unit; a face angle detector configured to detect a face angle that is an angle between the display unit and a face of a user; and a display controller configured to adjust an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle.

(5) A display method according to the present disclosure includes: detecting a face angle that is an angle between a display unit and a face of a user; detecting a gaze of the user; and adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle.

(6) A non-transitory computer-readable storage medium stores a display program causing a computer to execute: detecting a face angle that is an angle between a display unit and a face of a user; detecting a gaze of the user; and adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating an example of a display device according to a first embodiment;
- (2) FIG. 2 is a functional block diagram illustrating an example of the display device according to the first embodiment;
- (3) FIG. 3 is a diagram illustrating an example of a display method according to the first embodiment;
- (4) FIG. 4 is a diagram illustrating an example of the display method according to the first embodiment;
- (5) FIG. 5 is a diagram illustrating an example of the display method according to the first embodiment;
- (6) FIG. 6 is a diagram illustrating an example of the display method according to the first embodiment;
- (7) FIG. 7 is a diagram illustrating an example of the display method according to the first embodiment;
- (8) FIG. 8 is a flowchart illustrating a flow of operations of the display device according to the first embodiment;
- (9) FIG. 9 is a functional block diagram illustrating an example of a display device according to a second embodiment;
- (10) FIG. 10 is a diagram illustrating an example of a display method according to the second embodiment;
- (11) FIG. 11 is a diagram illustrating an example of the display method according to the second embodiment;
- (12) FIG. 12 is a diagram illustrating an example of the display method according to the second embodiment;
- (13) FIG. 13 is a diagram illustrating an example of a relationship between a subject of detection that is a gaze and an eyelid open/close motion of the user and content of control on the display in the case where the subject of detection is detected; and
- (14) FIG. 14 is a flowchart illustrating a flow of operations of the display device according to a second embodiment.

DETAILED DESCRIPTION

(15) An embodiment of a display device, a display method, and a display program according to the present disclosure will be described below based on the drawings. Note that the embodiment does not limit the invention. The components in the following embodiment contains ones that are replaceable easily by those skilled in the art or ones that are substantially the same.

First Embodiment

- (16) FIG. 1 is a schematic diagram illustrating an example of a display device **100** according to a first embodiment. FIG. 2 is a functional block diagram illustrating an example of the display device **100** according to the first embodiment. As illustrated in FIG. 1, the display device **100** includes a casing **10**, a display unit **20**, an imaging unit **30**, and a controller **40**. The display device **100**, for example, may be a portable information terminal device, such as a smartphone or a tablet.
- (17) The casing **10** holds the display unit **20**, the imaging unit **30**, and the controller **40**. The casing **10**, for example, may be provided with an acceleration sensor that is not illustrated in the drawings and that is capable of detecting walking by the user.
- (18) The display unit **20** is capable of displaying image information, such as a still image or a video, and text information, such as characters. For example, a display, such as a liquid crystal panel, is used as the display unit **20**. For the display unit **20**, a given reference direction that is used

when a face angle to be described below is detected is set. For example, when the display unit **20** is rectangular, the reference direction can be one of two sides orthogonal to each other.

(19) The imaging unit **30** is capable of capturing an image of the face and the eyes of the user. For example, a camera that is attached to the casing **10**, or the like, is taken as the imaging unit **30**. The imaging unit **30** generates imaging data by capturing an image of the face and the eyes of the user. The imaging unit **30** transmits the imaging data to the controller **40**. A plurality of the imaging units **30** may be provided.

(20) The controller **40** generally controls operations of the display device **100**. The controller **40** includes a processing device, such as a CPU (Central Processing Unit), and a storage device, such as a RAM (Random Access Memory) or a ROM (Read Only Memory). The controller **40** includes a face angle detector **41**, a gaze detector **42**, a display controller **43**, and a storage unit **44**.

(21) The face angle detector **41** detects a face angle that is an angle between the display unit **20** and the face of the user based on the imaging data that is generated by the imaging unit **30**. The face angle detector **41** detects a center line in a width direction in an area from the forehead of the user to the chin and detects an angle between the center line and a reference direction of the display unit **20** as a face angle. The face angle is a value presenting the relative angle between the reference direction of the display unit **20** and a center line direction of the face of the user. The face angle is a value presenting a difference in a position of rotation on an axial direction (rolling direction) of a center axis perpendicular to the display unit **20**. Because at least one of the reference direction of the display unit **20** and the center line direction of the face of the user changes and accordingly the face angle changes. Note that the face angle detector **41** may set a line connecting the centers of the right and left eyes of the user as the center line direction.

(22) The gaze detector **42** detects a gaze of the user based on the imaging data that is generated by the imaging unit **30**. The gaze detector **42**, for example, may be configured to detect a gaze based on a position of a pupil of the user and a position of a corneal reflection image or may be configured to detect a gaze based on a position of an inner corner of an eye of the user and a position of an iris.

(23) The display controller **43** controls a display operation in the display unit **20**. The display controller **43** adjusts an angle of display of the subject of display by rotating the subject of display on a reference position on the display unit **20** according to the result of detecting the face angle. When the face angle is at or larger than a threshold, the display controller **43** is able to rotate the subject of display. When the gaze of the user is detected on the display unit **20**, the display controller **43** sets, for the reference position, the position corresponding to the gaze.

(24) The display controller **43** determines whether the subject of display contains text information and, when it is determined that it contains text information, determines whether the gaze of the user is present in a part of display of the text information. When it is determined that the gaze of the user is present in a part of display of the text information, the display controller **43** sets, for the reference position, a center of the text information on which the gaze is present in the row direction.

(25) The display controller **43** is able to adjust the size of the subject of display so that all the text information that is displayed on the display unit **20** right before the rotation is displayed, and rotate the subject of display. When part of the text information is displayed in an enlarged manner right before the rotation, the display controller **43** is able to adjust the size of the subject of display so that the whole text information in the row direction is displayed, and rotate the subject of display.

(26) The storage unit **44** stores various types of information. The storage unit **44**, for example, has a storage, such as a hard disk drive or a solid state drive. An external storage medium, such as a removable disk, may be used as the storage unit **44**.

(27) The storage unit **44** stores a display program that causes a computer to execute a process of detecting a face angle that is an angle between the display unit **20** and the face of a user, a process of detecting a gaze of the user, and a process of adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result

of detecting the face angle. The storage unit **44** stores a display program that causes a computer to execute a process of detecting a face angle that is an angle between the display unit **20** and the face of a user, a process of detecting a gaze of the user, and a process of adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle and setting, for the reference position, a position corresponding to the gaze when the gaze of the user is detected on the display unit **20**.

(28) An example of a display method in the display device **100** described above will be described next. FIGS. **3** to **7** are diagrams illustrating an example of the display method according to the first embodiment. As illustrated in FIG. **3**, the display controller **43** displays a subject-of-display **21** on the display unit **20**. The user is able to view the subject-of-display **21** that is displayed on the display unit **20**. The subject-of-display **21**, for example, contains image information P, text information T, etc.

(29) The imaging unit **30** captures an image of the face and the eyes of the user who looks at the display unit **20**. The imaging unit **30** generates imaging data obtained by capturing the face and the eyes of the user and transmits the imaging data to the controller **40**. In the controller **40**, the face angle detector **41** detects a face angle of the user based on the imaging data. The gaze detector **42** detects a gaze of the user based on the imaging data.

(30) When, for example, the user lies on from the posture of sitting while looking at the display unit **20** of the display device **100**, the face angle between the user and the display unit **20** changes in some cases. When the face angle of the user that is detected by the face angle detector **41** exceeds a threshold, the display controller **43** adjusts the angle of display of the subject of display. As illustrated in FIG. **4**, the display controller **43** is able to set a center part of the image information P of the subject-of-display **21** for the reference position, cause the subject of display on the reference position by 90°, and display the image information P in an enlarged manner. The display controller **43** is not limited to a configuration in which an angle of rotation is 90° when the angle of display of the subject of display is adjusted and it may be an angle different from 90°, such as 30°, 60°, or 120°.

(31) When the subject-of-display **21** contains the text information T, the display controller **43** may perform the following process. In other words, the display controller **43** determines whether the subject of display contains the text information T and, when it is determined that it contains the text information T, determines whether a gaze EP of the user is present on a part of display of the text information. When it is determined that the gaze EP of the user is present on the part of display of the text information, the display controller **43** is able to set, for a reference position BP, a center of the text information on which the gaze EP is present in the row direction and rotate the subject-of-display **21** as illustrated in FIG. **5**. When the text information is displayed in multiple rows, the display controller **43** may set the reference position BP in a position that is shifted to a side of the next row with respect to the row in which the gaze EP is present. In this case, rotating the subject-of-display **21** makes it possible to display more text information that the user is going to read. The display controller **43** may set the reference position BP such that the reference position BP matches the gaze EP. Regardless whether the subject-of-display **21** contains the text information T, the display controller **43** may set, for the reference position, the position in which the gaze of the user is present on the display unit **20** when rotating the subject-of-display **21**.

(32) When rotating the subject-of-display **21**, the display controller **43** may adjust the size of the subject-of-display **21** so that the text information T that is displayed on the display unit **20** right before the rotation is all displayed as illustrated in FIG. **6**. In this case, the reference position BP that is set for the rotation need not be arranged at the center of the display unit **20**.

(33) When rotating the subject-of-display **21**, the display controller **43** determines whether part of the text information T is displayed in an enlarged manner on the display unit **20** as illustrated in FIG. **7** right before the rotation. When it is determined that part of the text information T is displayed in an enlarged manner on the display unit **20**, the display controller **43** may adjust the

size of the subject-of-display **21** such that the whole text information **T** in the row direction is displayed.

(34) FIG. **8** is a flowchart illustrating a flow of operations of the display device **100** according to the first embodiment. As illustrated in FIG. **8**, in the state where the subject of display is being displayed on the display unit **20**, the face angle detector **41** detects a face angle between the display unit **20** and the face of the user (step **S101**). Furthermore, the gaze detector **42** detects a gaze of the user (step **S102**).

(35) The display controller **43** determines whether the face angle exceeds the threshold (step **S103**) and, when it is determined that it does not exceed the threshold (No at step **S103**), maintains an angle of display of the subject of display (step **S104**).

(36) On the other hand, when it is determined that it exceeds the threshold (Yes at step **S103**), the display controller **43** determines whether the subject of display contains text information that is displayed in the row direction (step **S105**). When it is determined that it contains text information that is displayed in the row direction (Yes at step **S105**), the display controller **43** determines whether the gaze of the user is present in a part of display of the text information (step **S106**). When it is determined that the gaze of the user is present in the part of display of the text information (Yes at step **S106**), the display controller **43** sets, for a reference position, the center of the text information on which the gaze is present (step **S107**). When it is determined that it does not contain text information that is displayed in the row direction at step **S105** (Yes at step **S105**), or when it is determined that the gaze of the user is not present in the part of display of the text information (No at step **S106**), the display controller **43** sets a given position for a reference position (step **S109**). After setting the reference position, the display controller **43** rotates the subject of display on the reference position (step **S108**).

(37) As described above, the display device **100** according to the present embodiment includes the display unit **20**, the face angle detector **41** that detects a face angle that is an angle between the display unit **20** and the face of a user, the gaze detector **42** that detects a gaze of the user, and the display controller **43** that adjusts an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle and, when the gaze of the user is detected on the display unit **20**, sets a position corresponding to the gaze for the reference position.

(38) The display method according to the present embodiment includes detecting a face angle that is an angle between the display unit **20** and the face of a user, detecting a gaze of the user, and adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle and, when the gaze of the user is detected on the display unit **20**, setting a position corresponding to the gaze for the reference position.

(39) According to this configuration, in the case where the subject of display on the display unit **20** is rotated, when the gaze of the user is detected on the display unit **20**, a position corresponding to the gaze is set for the reference position and the subject of display is rotated. Accordingly, it is possible to display a part that the user was looking at right before the rotation in an appropriate position.

(40) In the display device **100** according to the present embodiment, the display controller **43** determines whether the subject of display contains text information that is displayed in a row direction, when it is determined that it contains the text information that is displayed in the row direction, determines whether the gaze of the user is present in a part of display of the text information, and, when it is determined that the gaze of the user is present in the part of display of the text information, sets, for the reference position, a center of the text information on which the gaze is present in the row direction. According to this configuration, it is possible to display a part of the text information that the user was looking at right before the rotation in an appropriate position.

(41) In the display device **100** according to the present embodiment, the display controller **43** adjusts a size of the subject of display so that the text information that is displayed on the display unit **20** right before the rotation is all displayed, and rotates the subject of display. According to the configuration, it is possible to reduce uncomfortableness to the user who was looking at the text information before and after the rotation.

(42) In the display device **100** according to the present embodiment, when part of the text information is displayed in an enlarged manner on the display unit **20** right before rotation, the display controller **43** may adjust the size of the subject of display such that the whole text information in the row direction is displayed. According to the configuration, it is possible to allow the user to sense the whole text information containing the part the user was looking at right before the rotation in the row direction.

Second Embodiment

(43) FIG. **9** is a functional block diagram illustrating an example of a display device **200** according to a second embodiment. As illustrated in FIG. **9**, the display device **200** includes the casing **10**, the display unit **20**, the imaging unit **30**, and a controller **50**. The configurations of the casing **10**, the display unit **20**, and the imaging unit **30** are the same as those of the display device **100** according to the first embodiment.

(44) In the present embodiment, the configuration of the controller **50** is different from that of the controller **40** of the display device **100** in the first embodiment. The difference will be described mainly below. The controller **50** includes a face angle detector **51**, a gaze eyelid-open/close detector **52**, a display controller **53**, and a storage unit **54**. The face angle detector **51** is the same as the face angle detector **41** of the first embodiment.

(45) Based on imaging data that is generated by the imaging unit **30**, the gaze eyelid-open/close detector **52** detects at least one of a gaze and an eyelid open/close motion of a user. The gaze eyelid-open/close detector **52**, for example, may be configured to detect a gaze based on a position of an iris of the user and a position of a corneal reflection image or may be configured to detect a gaze based on a position of an inner corner of an eye of the user and a position of an iris. The gaze eyelid-open/close detector **52** is able to detect a motion of an eyelid of the user based on image processing.

(46) The display controller **53** controls a display operation in the display unit **20**. As in the first embodiment, the display controller **53** adjusts an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting a face angle. The display controller **53** determines whether the user is in a given not-viewing state based on at least one of a gaze and an eyelid open/close motion of the user and, when it is determined that it is in the not-viewing state, does not rotate the subject of display. In the present embodiment, the not-viewing state includes at least one of a state in which the user has his/her eyelids closed and a state in which the gaze of the user is not present on the display unit **20**. For example, when the gaze eyelid-open/close detector **52** detects that the user has his/her eyelids closed, the display controller **53** is able to determine that the user is in the not-viewing state. For example, when the gaze eyelid-open/close detector **52** detects the gaze of the user in a direction deviating from the display unit **20**, the display controller **53** is able to determine that that user is in the not-viewing state. When it is detected that the not-viewing state ends, the display controller **53** adjusts an angle of display of the subject of display based on a face angle at the time when the not-viewing state ends.

(47) The display controller **53** is able to make at least one of a setting for an initial value of the angle of display of the subject of display and a change in the setting according to a period during which the user has his/her eyelids closed. The initial value of the angle of display is an angle of display of the subject of display that is displayed initially at the time when the operation of the display device **200** is started or restarted.

(48) The display controller **53** is able to change a mode of display on the display unit **20** based on

at least one of the gaze and the eyelid open/close motion of the user. For example, the position of the gaze on the display unit **20**, a displacement, etc., are listed as the gaze of the user. For example, the period during which the user has his/her eyelids closed, the number of blinks, etc., are listed as the eyelid open/close motion of the user. In the case where the display device **200** is additionally provided with another component, such as a sensor capable of detecting a walk of the user, or the like, by detecting an acceleration, or the like, the display controller **53** is able to change the mode of display by combining another component, such as the sensor, and at least one of the gaze and the eyelid open/close motion of the user.

(49) The storage unit **54** stores various types of information. The storage unit **54**, for example, has a storage, such as a hard disk drive or a solid state drive. An external storage medium, such as a removable disk, may be used as the storage unit **54**.

(50) The storage unit **54** stores a display program that causes a computer to execute a process of detecting a face angle that is an angle between the display unit **20** and the face of a user, a process of detecting a gaze of the user, and a process of adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle. The storage unit **54** stores a display program that causes a computer to execute a process of detecting a face angle that is an angle between the display unit **20** and the face of a user, a process of detecting at least one of a gaze and an eyelid open/close motion of the user, and a process of adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle, determining whether the user is in a given not-viewing state based on at least one of the gaze and an eyelid open/close motion of the user and, when it is determined that it is in the not-viewing state, not rotating the subject of display.

(51) An example of a display method in the display device **200** described above will be described next. FIGS. **10** to **12** are diagrams illustrating an example of the display method according to the second embodiment. The display controller **53** displays the subject-of-display **21** on the display unit **20**. The user is able to view the subject-of-display **21** that is displayed on the display unit **20**.

(52) The imaging unit **30** captures an image of the face and the eyes of the user who looks at the display unit **20**. The imaging unit **30** generates imaging data obtained by capturing the face and the eyes of the user and transmits the imaging data to the controller **50**. In the controller **50**, the face angle detector **51** detects a face angle of the user based on the imaging data. The gaze eyelid-open/close detector **52** detects a gaze or the eyelid open/close motion of the user based on the imaging data.

(53) When the face angle of the user that is detected by the face angle detector **51** exceeds a threshold, the display controller **53** adjusts the angle of display of the subject of display. When the face angle changes, the display controller **53** determines whether the user is in the not-viewing state. When it is detected that the user has his/her eyelids opened and when it is detected that the gaze of the user is present on the display unit **20**, the display controller **53** determines that the user is not in the not-viewing state. In this case, the display controller **53** adjusts the angle of display by rotating the subject-of-display **21** displayed on the display unit **20** as illustrated in FIG. **10**.

(54) On the other hand, when it is detected that the user has his/her eyelids closed, or when it is detected that the gaze of the user is not present on the display unit **20**, the display controller **53** determines that the user is in the not-viewing state. In this case, the display controller **53** prevents the subject-of-display **21** displayed on the display unit **20** from rotating as illustrated in FIG. **11**. Thereafter, when it is detected that the user opens the eyelids late and when it is detected that the gaze of the user is present on the display unit **20**, the display controller **53** determines that the not-viewing state of the user ends. In this case, as illustrated in FIG. **12**, based on the face angle at the time when it is determined that the not-viewing state ends, the display controller **53** adjusts the angle of display of the subject-of-display **21** displayed on the display unit **20**.

(55) FIG. **13** is a diagram illustrating an example of a relationship between a subject of detection

that is the gaze and the eyelid open/close motion of the user and content of control on the display in the case where the subject of detection is detected. As illustrated in FIG. 13, the display controller 53 is able to make at least one of a setting for an initial value of the angle of display of the subject of display and a change in the setting according to a period during which the user has his/her eyelids closed. When a motion of opening the eyelids after closing the eyelids for a first period (for example, 1 second) by the user is detected, the display controller 53 is able to set an angle of display of the subject of display at the time when the user has the eyelids opened for an initial value of the angle of display on the display unit 20. After the user sets the initial value of the angle of display, when a motion of opening the eyelids is detected after the eyelids are closed for a second period (for example, two seconds), the display controller 53 is able to cancel the setting of the initial value of the angle of display.

(56) As illustrated in FIG. 13, based on at least one of the gaze and the eyelid open/close motion of the user, the display controller 53 is able to change the mode of display of the display unit 20. For example, when the gaze of the user is detected on an upper part of the display unit 20 and a blink motion is detected for a first number of times (for example, twice), the display controller 53 is able to scroll up the subject of display by half the screen of the display unit 20. When the gaze of the user is detected on a lower part of the display unit 20 and the blink motion is detected for the first number of times (for example, twice), the display controller 53 is able to scroll down the subject of display by half the screen of the display unit 20.

(57) As illustrated in FIG. 13, in the case where the gaze of the user is detected on the upper part or the lower part of the display unit 20, when the blink motion is detected for a second number of times (for example, third times) more than the first number of times, the display controller 53 is able to scroll up or down the subject of display by one screen of the display unit 20 in each case. As described above, the display controller 53 is able to set a direction in which the subject of display is scrolled based on the position of the gaze on the display unit 20 and set an amount of scrolling based on the number of blinks.

(58) As illustrated in FIG. 13, in the case where the gaze of the user is detected on the upper part or the lower part of the display unit 20, when the blink motion is detected for the aforementioned given number of times (for example, twice), the display controller 53 may make a smooth scroll up or down of the subject of display. When it is detected that the user closes the eyelids for a given period (for example, 1 second) in the middle of the smooth scroll, the display controller 53 may stop the scroll.

(59) As illustrated in FIG. 13, the display controller 53 may determine whether text information is contained in the subject of display and, when it is determined that text information is contained, scroll the subject of display in accordance with an action of reading the text information line by line by the user. For example, when the gaze of the user shifts from a line to the next line, the display controller 53 may scroll the subject of display in accordance with the shift of the gaze of the user with respect to a column direction orthogonal to the row direction.

(60) As illustrated in FIG. 13, the display controller 53 is able to detect the number of blinks by the user and a length of time of a blink and, when the number of blinks per unit of time exceeds a threshold or the length of time per blink is longer than a given threshold, determine that the eyes of the user fatigue. When it is determined that the eyes of the user fatigue, the display controller 53 is able to adjust illumination of the display unit 20. For example, when the given threshold is exceeded or it is longer than the given threshold, the display controller 53 is able to reduce the luminance of the display unit 20 and, when it returns to be at or smaller than the given threshold, return the luminance of the display unit 20 to the original luminance, thereby adjusting on the luminance.

(61) As illustrated in FIG. 13, in the case where a sensor capable of detecting a walk of the user, or the like, by detecting vibrations of the casing 10, an acceleration, or the like, is provided additionally, when a walk by the user is detected, the display controller 53 is able to determine

whether the gaze of the user is present on the display unit **20**. In the case where it is determined that the gaze of the user is present on the display unit **20** when a walk by the user is detected, the display controller **53** is able to display alert information on looking at the display unit **20** while walking and turn off the display of the display unit **20**.

(62) FIG. **14** is a flowchart illustrating a flow of operations of the display device **100**. As illustrated in FIG. **14**, in the state where the subject of display is being displayed on the display unit **20**, the face angle detector **51** detects a face angle between the display unit **20** and the face of a user (step **S201**). The gaze eyelid-open/close detector **52** detects at least one of a gaze and an eyelid open/close motion of the user (step **S202**).

(63) The display controller **53** determines whether the face angle exceeds the threshold (step **S203**) and, when it is determined that it does not exceed the threshold (No at step **S203**), maintains an angle of display of the subject of display (step **S204**).

(64) On the other hand, when it is determined that it exceeds the threshold (Yes at step **S203**), the display controller **53** determines whether the user is in the given not-viewing state (step **S205**). When it is determined that the user is not in the given not-viewing state (No at step **S205**), the subject of display is rotated on a reference position and thereby the angle of display of the subject of display is adjusted (step **S206**).

(65) When it is determined that the user is in the given not-viewing state (Yes at step **S205**), the display controller **53** prevents the subject of display from rotating (step **S207**). Thereafter, it is determined whether the not-viewing state of the user ends (step **S208**) and, when it is determined that the not-viewing state does not end (No at step **S208**), the operation at step **S208** is performed repeatedly and thereby the state where the subject of display is prevented from rotating is maintained. When it is determined that the not-viewing state ends (Yes at step **S208**), the display controller **53** adjusts the angle of display of the subject of display based on the face angle at the time of the determination. In other words, the display controller **53** determines whether the face angle exceeds the threshold at the time of the determination (step **S209**) and, when it is determined that it does not exceed the threshold (No at step **S209**), maintains the angle of display of the subject of display (step **S204**). On the other hand, when it is determined that it exceeds the threshold (Yes at step **S209**), the subject of display is rotated on the reference position and thereby the angle of display of the subject of display is adjusted (step **S206**).

(66) As described above, the display device **200** according to the present embodiment includes the display unit **20**, the face angle detector **51** that detects a face angle that is an angle between the display unit **20** and the face of a user, the gaze eyelid-open/close detector **52** that detects at least one of a gaze and an eyelid open/close motion of the user, and the display controller **53** that adjusts an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle, determines whether the user is in the given not-viewing state based on at least one of the gaze and the eyelid open/close motion of the user and, when it is determined that it is in the given not-viewing state, prevents the subject of display from rotating.

(67) The display method according to the present embodiment includes detecting a face angle that is an angle between the display unit **20** and the face of a user, detecting at least one of a gaze and an eyelid open/close motion of the user, and adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle, determining whether the user is in the given not-viewing state based on at least one of the gaze and the eyelid open/close motion of the user and, when it is determined that it is in the given not-viewing state, preventing the subject of display from rotating.

(68) According to this configuration, in the case where it is determined that the user is in the not-viewing state, the subject of display does not rotate. Accordingly, it is possible to reduce uncomfortableness given to the user when the subject of display on the display unit **20** is rotated.

(69) In the display device **200** according to the present embodiment, when it is detected that the

not-viewing state ends, the display controller **53** adjusts the angle of display of the subject of display based on the face angle at the time when the not-viewing state ends. According to this configuration, when the not-viewing state ends, because the angle of display of the subject of display is adjusted according to the face angle at that time, it is possible to appropriately set the angle of display of the subject of display at timing when the user looks at the display unit **20**.

(70) In the display device **200** according to the present embodiment, the gaze eyelid-open/close detector **52** detects the eyelid open/close motion of the user and the display controller **53** makes at least one of a setting for an initial value of the angle of display of the subject of display and a change in the setting according to a period during which the user has his/her eyelids closed.

According to this configuration, it is possible to make a setting for an initial value of the angle of display of the subject of display and a change in the setting by a simple operation.

(71) In the display device **200** according to the present embodiment, the gaze eyelid-open/close detector **52** detects both the gaze and the eyelid open/close motion of the user and the display controller **53** changes the mode of display of the display unit **20** based on at least one of the gaze and the eyelid open/close motion of the user. According to this configuration, it is possible to appropriately change the mode of display of the display unit **20** by a simple operation.

(72) As described above, according to the mode of the present disclosure, the display device **100**, **200** including the display unit **20**, the face angle detector **41**, **51** that detects a face angle that is an angle between the display unit **20** and the face of a user, and the display controller **43**, **54** that adjusts an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle is provided.

(73) According to the mode of the present disclosure, a display method including detecting a face angle that is an angle between the display unit **20** and the face of a user, detecting a gaze of the user, and adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle is provided.

(74) According to the mode of the present disclosure, a display program that causes a computer to execute a process of detecting a face angle that is an angle between the display unit **20** and the face of a user, a process of detecting a gaze of the user, and a process of adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit **20** according to a result of detecting the face angle is provided.

(75) According to the mode of the disclosure, because the face angle that is the angle between the display unit **20** and the face of the user is detected and the angle of display of the subject of display is adjusted by rotating the subject of display on the reference position on the display unit **20** according to the result of detecting the face angle, it is possible to appropriately perform the operation of rotating the subject-of-display **21** displayed on the display unit **20** according to the face angle.

(76) The technical field of the disclosure is not limited to the above-described embodiments and it is possible to add changes without departing from the scope of the present disclosure. For example, in the above-described embodiment, the configuration in which the imaging unit is provided in the casing **10** is exemplified and described; however, the configuration is not limited to this. The imaging unit **30** may be provided independently of the casing **10**, that is, the display device **100**, **200**.

(77) The display program described above may be provided by being stored in a non-transitory computer-readable storage medium, or may be provided via a network such as the Internet. Examples of the computer-readable storage medium include optical discs such as a digital versatile disc (DVD) and a compact disc (CD), and other types of storage devices such as a hard disk and a semiconductor memory.

(78) According to the disclosure, it is possible to perform the operation of rotating a subject of display that is displayed on the display unit.

(79) It is possible to use the display device, the display method, and the display program according to the disclosure for, for example, a processing device, such as a computer, etc.

(80) Although the present disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A display device comprising: a display unit; a face angle detector configured to detect a face angle that is an angle between the display unit and a face of a user; a gaze detector configured to detect a gaze of the user; and a display controller configured to adjust an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle, and when the gaze of the user is detected on the display unit, set a position corresponding to the gaze for the reference position.
2. The display device according to claim 1, wherein the display controller is configured to determine whether the subject of display contains text information that is displayed in a row direction, when the subject of display is determined to contain the text information displayed in the row direction, determine whether the gaze of the user is present in a part of display of the text information, and when the gaze of the user is determined to be present in the part of display of the text information, set a center of the text information on which the gaze is present in the row direction for the reference position.
3. The display device according to claim 2, wherein the display controller is configured to adjust a size of the subject of display so that the text information that is displayed on the display unit right before rotation is all displayed, and rotate the subject of display.
4. The display device according to claim 2, wherein, when part of the text information is displayed in an enlarged manner on the display unit right before rotation, the display controller is configured to adjust the size of the subject of display so that the whole text information in the row direction is displayed, and rotate the subject of display.
5. The display device according to claim 1, further comprising a gaze eyelid-open/close detector configured to detect at least one of a gaze and an eyelid open/close motion of the user, wherein the display controller is configured to determine whether the user is in a not-viewing state based on at least one of the gaze and the eyelid open/close motion of the user, and when the user is determined to be in the not-viewing state, prevent the subject of display from rotating.
6. The display device according to claim 5, wherein, when the not-viewing state is detected to have ended, the display controller is configured to adjust the angle of display of the subject of display based on the face angle at a time when the not-viewing state ended.
7. The display device according to claim 5, wherein the gaze eyelid-open/close detector is configured to detect the eyelid open/close motion of the user, and the display controller is configured to make at least one of a setting for an initial value of the angle of display of the subject of display and a change in the setting according to a period during which the user has eyelids closed.
8. The display device according to claim 5, wherein the gaze eyelid-open/close detector is configured to detect both the gaze and the eyelid open/close motion of the user, and the display controller is configured to change a mode of display of the display unit based on at least one of the gaze and the eyelid open/close motion of the user.
9. A display method comprising: detecting a face angle that is an angle between a display unit and a face of a user; detecting a gaze of the user; adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle; and when the gaze of the user is detected on the display unit, setting a

position corresponding to the gaze for the reference position.

10. A non-transitory computer-readable storage medium storing a display program causing a computer to execute: detecting a face angle that is an angle between a display unit and a face of a user; detecting a gaze of the user; adjusting an angle of display of a subject of display by rotating the subject of display on a reference position on the display unit according to a result of detecting the face angle; and when the gaze of the user is detected on the display unit, setting a position corresponding to the gaze for the reference position.
